CAPACITANCE

19.1 Capacitors and capacitance

Candidates should be able to:

1 define capacitance, as applied to both isolated spherical conductors and to parallel plate capacitors

2 recall and use $C = Q / V$

3 derive, using $C = Q / V$, formulae for the combined capacitance of capacitors in series and in parallel

4 use the capacitance formulae for capacitors in series and in parallel

19.2 Energy stored in a capacitor

Candidates should be able to:

1 determine the electric potential energy stored in a capacitor from the area under the potential–charge graph

2 recall and use $W = \frac{1}{2}QV = \frac{1}{2}CV^2$

19.3 Discharging a capacitor

Candidates should be able to:

1 analyse graphs of the variation with time of potential difference, charge and current for a capacitor discharging through a resistor

2 recall and use $\tau = RC$ for the time constant for a capacitor discharging through a resistor

3 use equations of the form $x = x_0 \, e^{-(t/RC)}$ where x could represent current, charge or potential difference for a capacitor discharging through a resistor

1. O/N/07/Q5

(a) State one function of capacitors in simple circuits.

....

....[1]

(b) A capacitor is charged to a potential difference of 15 V and then connected in series with a switch, a resistor of resistance 12 kΩ and a sensitive ammeter, as shown in Fig. 5.1.

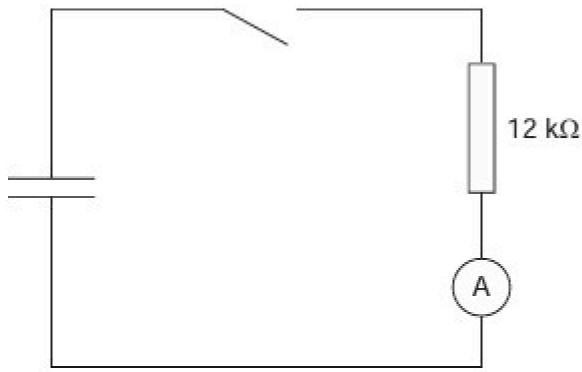
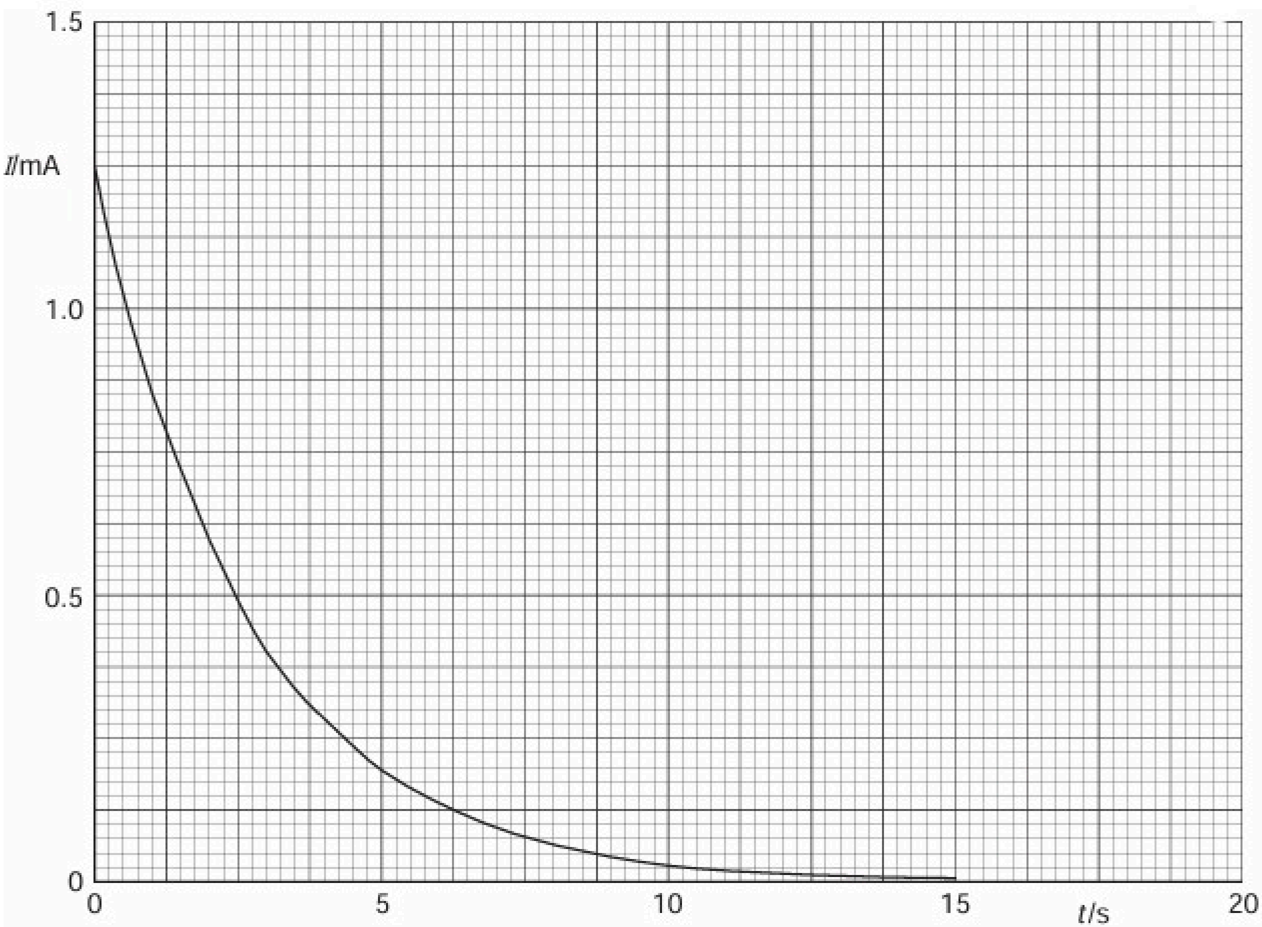


Fig. 5.1

The switch is closed and the variation with time t of the current I in the circuit is shown in Fig. 5.2.



(i) State the relation between the current in a circuit and the charge that passes a point in the circuit.

....

....[1]

(ii) The area below the graph line of Fig. 5.2 represents charge.
Use Fig. 5.2 to determine the initial charge stored in the capacitor.

charge = ... μC [4]

(iii) Initially, the potential difference across the capacitor was 15 V.
Calculate the capacitance of the capacitor.

capacitance = ... μF [2]

(c) The capacitor in **(b)** discharges one half of its initial energy. Calculate the new potential difference across the capacitor.

potential difference = ...V [3]

2. M/J/08/Q5

A capacitor C is charged using a supply of e.m.f. 8.0V. It is then discharged through a resistor R.
The circuit is shown in Fig. 5.1.

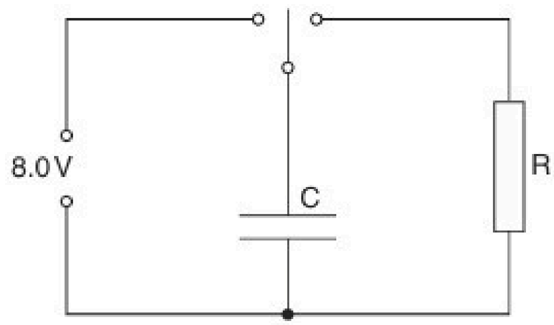


Fig. 5.1

The variation with time t of the potential difference V across the resistor R during the discharge of the capacitor is shown in Fig. 5.2.

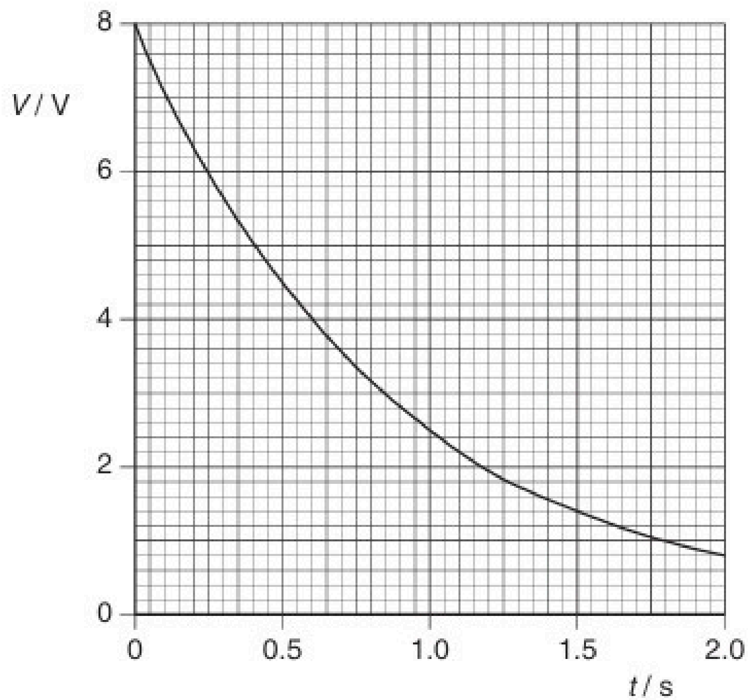


Fig. 5.2

(a) During the first 1.0s of the discharge of the capacitor, 0.13J of energy is transferred to the resistor R.
Show that the capacitance of the capacitor C is 4500 μF.

[3]

(b) Some capacitors, each of capacitance 4500 μF with a maximum working voltage of 6V, are available.

Draw an arrangement of these capacitors that could provide a total capacitance of 4500 μF for use in the circuit of Fig. 5.1.

[2]

3. M/J/09/Q5

A solid metal sphere, of radius r, is insulated from its surroundings. The sphere has charge $+Q$.
This charge is on the surface of the sphere but it may be considered to be a point charge at its centre, as illustrated in Fig. 5.1.

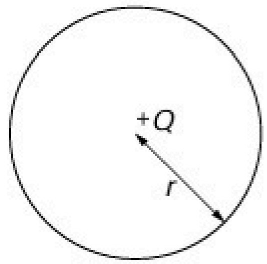


Fig. 5.1

(a) (i) Define *capacitance*.

....

.... [1]

(ii) Show that the capacitance C of the sphere is given by the expression

$$C = 4\pi\varepsilon_0 r.$$

[1]

(b) The sphere has radius 36 cm.
Determine, for this sphere,

(i) the capacitance,

capacitance = ... F [1]

(ii) the charge required to raise the potential of the sphere from zero to 7.0×10^5 V.

charge = ... C [1]

(c) Suggest why your calculations in **(b)** for the metal sphere would not apply to a plastic sphere.

....

....

....

.... [3]

(d) A spark suddenly connects the metal sphere in **(b)** to the Earth, causing the potential of the sphere to be reduced from 7.0×10^5 V to 2.5×10^5 V.

Calculate the energy dissipated in the spark.

energy = ... J [3]

4. O/N/09/42/Q4

(a) Define *capacitance*.

....

.... [1]

(b) An isolated metal sphere of radius R has a charge $+Q$ on it.

The charge may be considered to act as a point charge at the centre of the sphere.

Show that the capacitance C of the sphere is given by the expression

$$C = 4\pi\varepsilon_0 R$$

where ε_0 is the permittivity of free space.

[1]

(c) In order to investigate electrical discharges (lightning) in a laboratory, an isolated metal sphere of radius 63 cm is charged to a potential of 1.2×10^6 V.

At this potential, there is an electrical discharge in which the sphere loses 75% of its energy.

Calculate

(i) the capacitance of the sphere, stating the unit in which it is measured,

capacitance = ... [3]

(ii) the potential of the sphere after the discharge has taken place.

potential = ... V [3]

5. M/J/10/42/Q5

(a) State two functions of capacitors in electrical circuits.

1. ....

2. ....

[2]

(b) Three capacitors, each marked '30 μF, 6V max', are arranged as shown in Fig. 5.1.

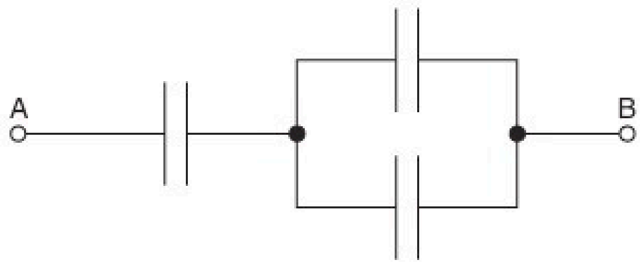


Fig. 5.1

Determine, for the arrangement shown in Fig. 5.1,

(i) the total capacitance,

capacitance = ... μF [2]

(ii) the maximum potential difference that can safely be applied between points A and B.

potential difference = ... V [2]

(c) A capacitor of capacitance 4700 μF is charged to a potential difference of 18 V. It is then partially discharged through a resistor. The potential difference is reduced to 12 V. Calculate the energy dissipated in the resistor during the discharge.

energy = ... J [3]

6. O/N/10/41/Q4

(a) Define *capacitance*.

....

.... [1]

(b) An isolated metal sphere has a radius r. When charged to a potential V, the charge on the sphere is q.
The charge may be considered to act as a point charge at the centre of the sphere.

(i) State an expression, in terms of r and q, for the potential V of the sphere.

.... [1]

(ii) This isolated sphere has capacitance. Use your answers in **(a)** and **(b)(i)** to show that the capacitance of the sphere is proportional to its radius.

[1]

(c) The sphere in **(b)** has a capacitance of 6.8 pF and is charged to a potential of 220 V.

Calculate

(i) the radius of the sphere,

radius = ... m [3]

(ii) the charge, in coulomb, on the sphere.

charge = ... C [1]

(d) A second uncharged metal sphere is brought up to the sphere in **(c)** so that they touch. The combined capacitance of the two spheres is 18 pF.

Calculate

(i) the potential of the two spheres,

potential = ... V [1]

(ii) the change in the total energy stored on the spheres when they touch.

change = ... J [3]

7. O/N/10/43/Q4

(a) **(i)** State what is meant by *electric potential* at a point.

....

....

.... [2]

(ii) Define *capacitance*.

....

.... [1]

(b) The variation of the potential V of an isolated metal sphere with charge Q on its surface is shown in Fig. 4.1.

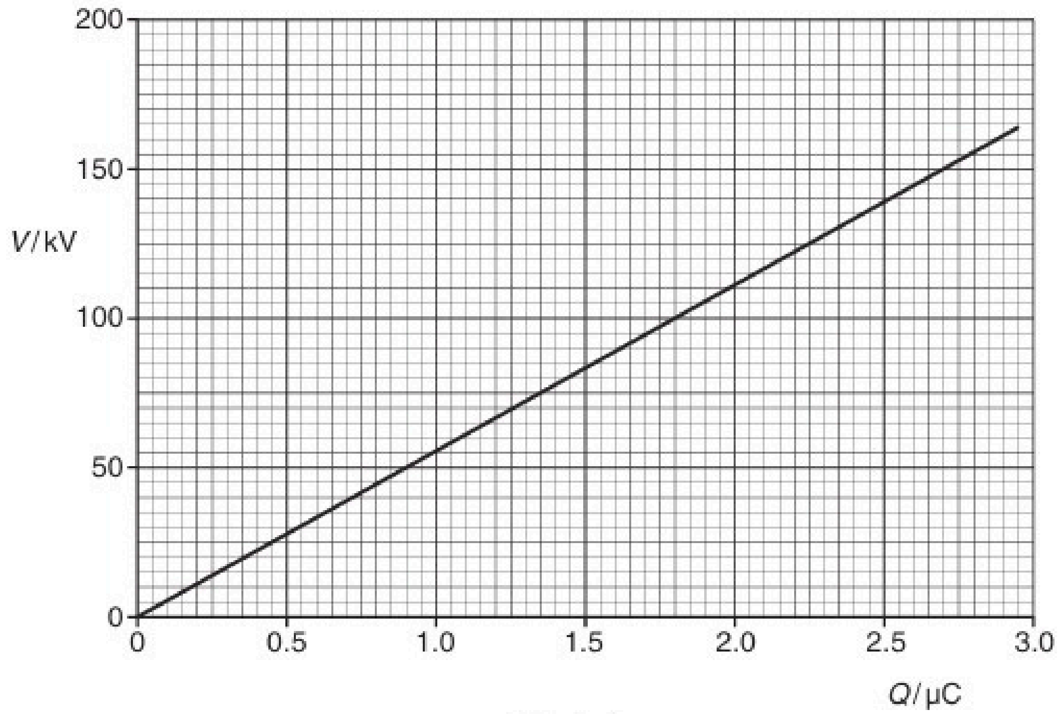


Fig. 4.1

An isolated metal sphere has capacitance.

Use Fig. 4.1 to determine

(i) the capacitance of the sphere,

capacitance = ... F [2]

(ii) the electric potential energy stored on the sphere when charged to a potential of 150 kV.

energy = ... J [2]

(c) A spark reduces the potential of the sphere from 150 kV to 75 kV.
Calculate the energy lost from the sphere.

energy = ... J [2]

8. M/J/11/42/Q3

A capacitor consists of two metal plates separated by an insulator, as shown in Fig. 3.1.

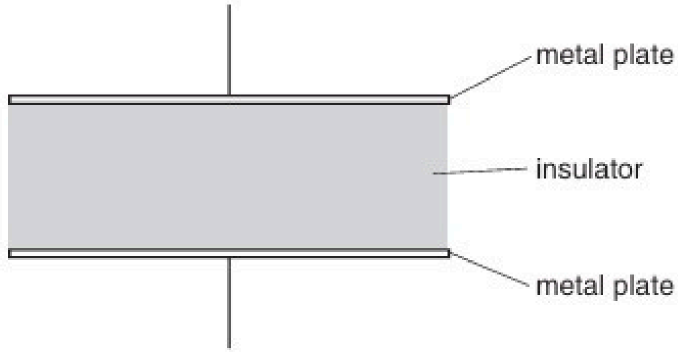


Fig. 3.1

The potential difference between the plates is V. The variation with V of the magnitude of the charge Q on one plate is shown in Fig. 3.2.

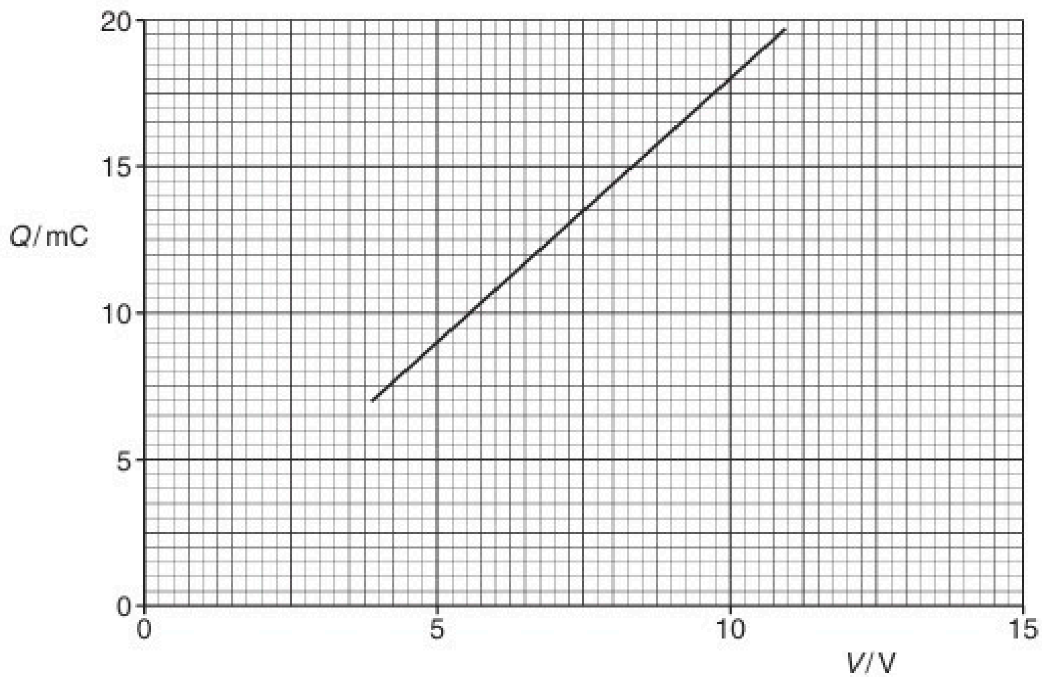


Fig. 3.2

(a) Explain why the capacitor stores energy but not charge.

....

....

....

.... [3]

(b) Use Fig. 3.2 to determine

(i) the capacitance of the capacitor,

capacitance = ... μF [2]

(ii) the loss in energy stored in the capacitor when the potential difference *V* is reduced from 10.0V to 7.5V.

energy = ... mJ [2]

(c) Three capacitors X, Y and Z, each of capacitance 10 μF, are connected as shown in Fig. 3.3.

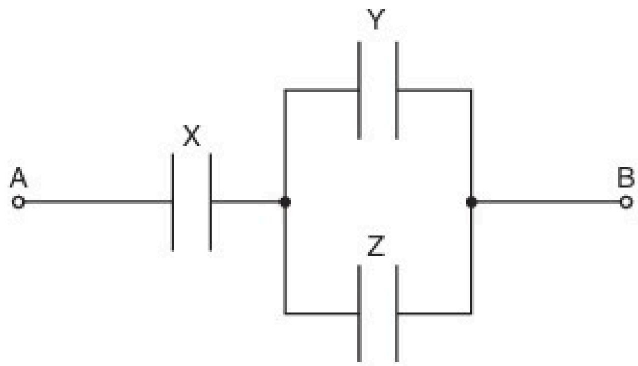


Fig. 3.3

Initially, the capacitors are uncharged.
A potential difference of 12 V is applied between points A and B.
Determine the magnitude of the charge on one plate of capacitor X.

charge = ... μC [3]

9. O/N/11/43/Q4

(a) State two functions of capacitors in electrical circuits.

1. ....

....

2. ....

.... [2]

(b) Three uncharged capacitors of capacitance C_1, C_2 and C_3 are connected in series, as shown in Fig. 4.1.

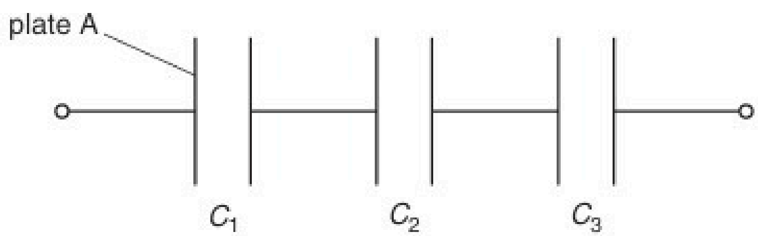


Fig. 4.1

A charge of $+Q$ is put on plate A of the capacitor of capacitance C_1.

(i) State and explain the charges that will be observed on the other plates of the capacitors.
You may draw on Fig. 4.1 if you wish.

....

....

.... [2]

(ii) Use your answer in **(i)** to derive an expression for the combined capacitance of the capacitors.

[2]

(c) A capacitor of capacitance 12 μF is charged using a battery of e.m.f. 9.0 V, as shown in Fig. 4.2.

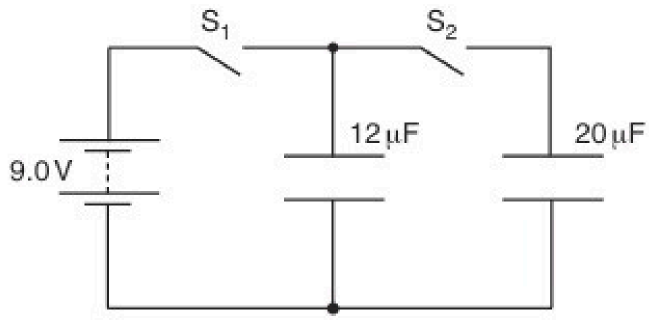


Fig. 4.2

Switch S_1 is closed and switch S_2 is open.

(i) The capacitor is now disconnected from the battery by opening S_1.
Calculate the energy stored in the capacitor.

energy = ... J [2]

(ii) The 12 μF capacitor is now connected to an uncharged capacitor of capacitance 20 μF by closing S_2. Switch S_1 remains open.
The total energy now stored in the two capacitors is 1.82×10^{-4} J.

Suggest why this value is different from your answer in **(i).**

....

.... [1]

10. O/N/12/41/Q5

(a) **(i)** Define *capacitance.*

....

....[1]

(ii) A capacitor is made of two metal plates, insulated from one another, as shown in Fig. 5.1.

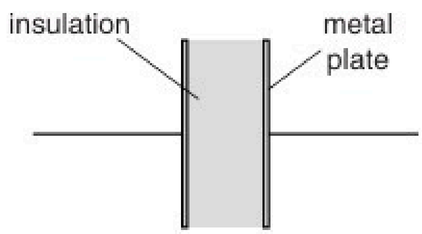


Fig. 5.1

Explain why the capacitor is said to store energy but not charge.

....

....

....

....

....[4]

(b) Three uncharged capacitors X, Y and Z, each of capacitance 12 μF, are connected as shown in Fig. 5.2.

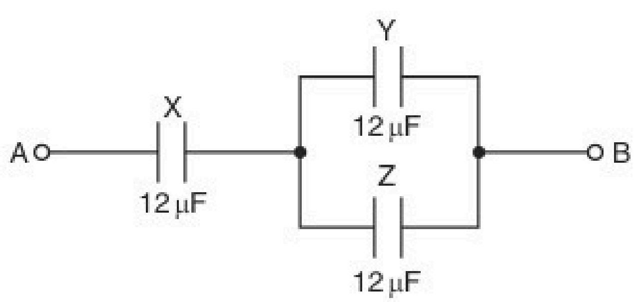


Fig. 5.2

A potential difference of 9.0 V is applied between points A and B.

(i) Calculate the combined capacitance of the capacitors X, Y and Z.

capacitance = ... μF [2]

(ii) Explain why, when the potential difference of 9.0 V is applied, the charge on one plate of capacitor X is 72 μC.

....

....

....[2]

(iii) Determine

1. the potential difference across capacitor X,

potential difference = ... V [1]

2. the charge on one plate of capacitor Y.

charge = ... μC [2]

11. M/J/13/42/Q4

(a) An insulated metal sphere of radius R is situated in a vacuum. The charge q on the sphere may be considered to be a point charge at the centre of the sphere.

(i) State a formula, in terms of R and q, for the potential V on the surface of the sphere.

.... [1]

(ii) Define capacitance and hence show that the capacitance C of the sphere is given by the expression

$$C = 4\pi\varepsilon_0 R.$$

[1]

(b) An isolated metal sphere has radius 45 cm.

(i) Use the expression in **(a)(ii)** to calculate the capacitance, in picofarad, of the sphere.

capacitance = ... pF [2]

(ii) The sphere is charged to a potential of 9.0×10^5 V.
A spark occurs, partially discharging the sphere so that its potential is reduced to 3.6×10^5 V.

Determine the energy of the spark.

[3]

12. O/N/13/43/Q4

(a) State two functions of capacitors connected in electrical circuits.

1. ....

....

2. ....

.... [2]

(b) Three capacitors are connected in parallel to a power supply as shown in Fig. 4.1.

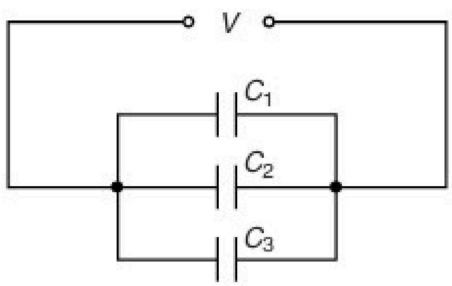


Fig. 4.1

The capacitors have capacitances C_1, C_2 and C_3. The power supply provides a potential difference V.

(i) Explain why the charge on the positive plate of each capacitor is different.

....

....

.... [1]

(ii) Use your answer in **(i)** to show that the combined capacitance C of the three capacitors is given by the expression

$$C = C_1 + C_2 + C_3.$$

[2]

(c) A student has available three capacitors, each of capacitance 12 μF.
Draw circuit diagrams, one in each case, to show how the student connects the three capacitors to provide a combined capacitance of

(i) 8 μF,

[1]

(ii) 18 μF.

[1]

13. M/J/14/41/Q6

An uncharged capacitor is connected in series with a battery, a switch and a resistor, as shown in Fig. 6.1.

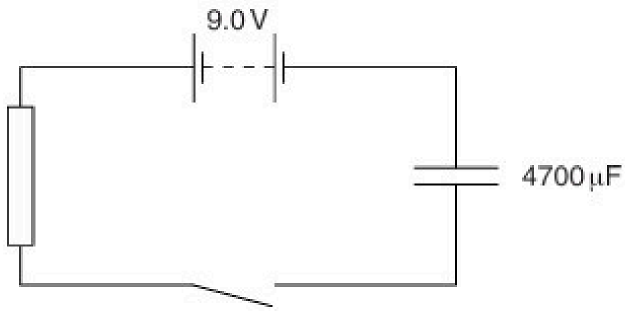


Fig. 6.1

The battery has e.m.f. 9.0 V and negligible internal resistance. The capacitance of the capacitor is 4700 μF.
The switch is closed at time $t = 0$.
During the time interval $t = 0$ to $t = 4.0$ s, the charge passing through the resistor is 22 mC.

(a) (i) Calculate the energy transfer in the battery during the time interval $t = 0$ to $t = 4.0$ s.

energy transfer = ... J [2]

(ii) Determine, for the capacitor at time $t = 4.0$ s,

1. the potential difference V across the capacitor,

V = ... V [2]

2. the energy stored in the capacitor.

energy = ... J [2]

(b) Suggest why your answers in **(a)(i)** and **(a)(ii) part 2** are different.

....

.... [1]

_14. O/N/14/43/Q6_

Three capacitors, each of capacitance 48 μF, are connected as shown in Fig. 6.1.

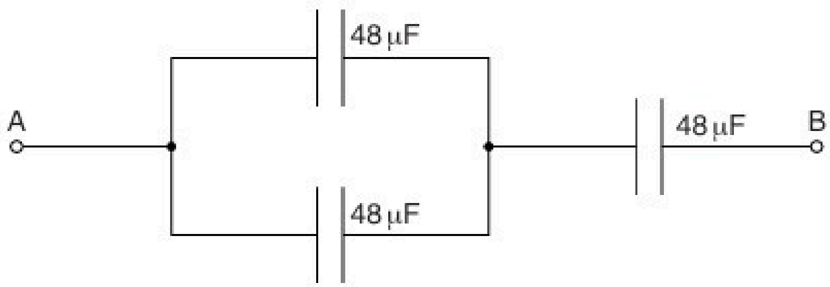


Fig. 6.1

(a) Calculate the total capacitance between points A and B.

capacitance = ... μF [2]

(b) The maximum safe potential difference that can be applied across any one capacitor is 6 V.

Determine the maximum safe potential difference that can be applied between points A and B.

potential difference = ... V [2]

15. *F/M/16/42/Q7*

(a) Define *capacitance*.

....

....[1]

(b) Three capacitors of capacitances C_1, C_2 and C_3 are initially uncharged. They are then connected in series to a battery, as shown in Fig. 7.1.

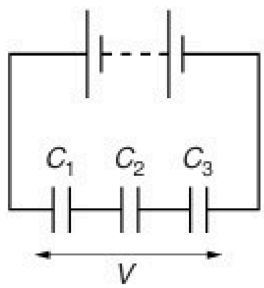


Fig. 7.1

The battery applies a potential difference V across the three capacitors.

Show that the combined capacitance C of the capacitors is given by

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}.$$

[2]

(c) A battery of e.m.f. 12V and negligible internal resistance is connected to a network of two capacitors and a resistor, as shown in Fig. 7.2.

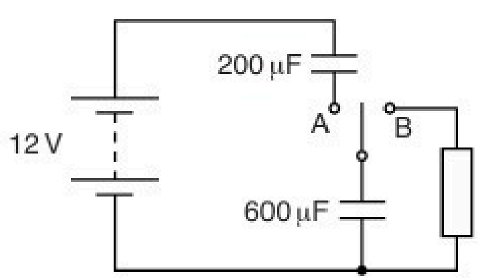


Fig. 7.2

The capacitors have capacitances of 200 μF and 600 μF. The switch has two positions, A and B.

(i) The switch is moved to position A.

Calculate

1. the combined capacitance of the two capacitors,

combined capacitance = ... μF [1]

2. the charge on the 600 μF capacitor,

charge = ... C [1]

3. the potential difference across the 600 μF capacitor.

potential difference = ... V [1]

(ii) The switch is now moved from position A to position B.

Calculate the potential difference across the 600 μF capacitor when it has discharged 50% of its initial energy.

potential difference = ... V [3]

[Total: 9]

16. M/J/16/42/Q7

(a) State two uses of capacitors in electrical circuits, other than for the smoothing of direct current.

1. ....

2. ....
[2]

(b) The combined capacitance between terminals A and B of the arrangement shown in Fig. 7.1 is 4.0 μF.

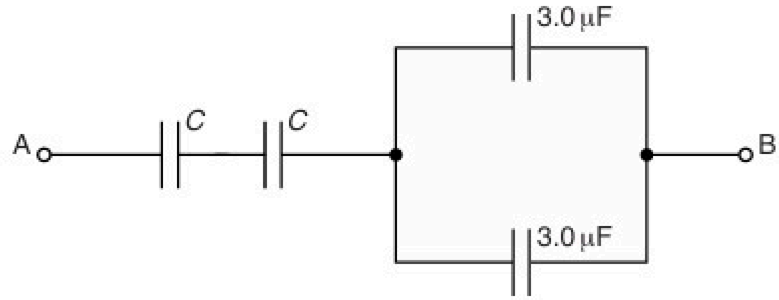


Fig. 7.1

Two capacitors each have capacitance C and the remaining capacitors each have capacitance 3.0 μF.

The potential difference (p.d.) between terminals A and B is 12 V.

(i) Determine the capacitance C.

C = ... μF [2]

(ii) Calculate the magnitude of the total positive charge transferred to the arrangement.

charge = ... μC [2]

(iii) Use your answer in **(ii)** to state the magnitude of the charge on one plate of

1. a capacitor of capacitance *C*,

charge = ... μC

2. a capacitor of capacitance 3.0 μF.

charge = ... μC
[2]

[Total: 8]

17. O/N/16/42/Q7

(a) **(i)** Define *capacitance*.

....

....[1]

(ii) Use the expression for the electric potential due to a point charge to show that an isolated metal sphere of diameter 25 cm has a capacitance of 1.4×10^{-11} F.

[2]

(b) Three capacitors of capacitances 2.0 μF, 3.0 μF and 4.0 μF are connected as shown in Fig. 7.1 to a battery of e.m.f. 9.0 V.

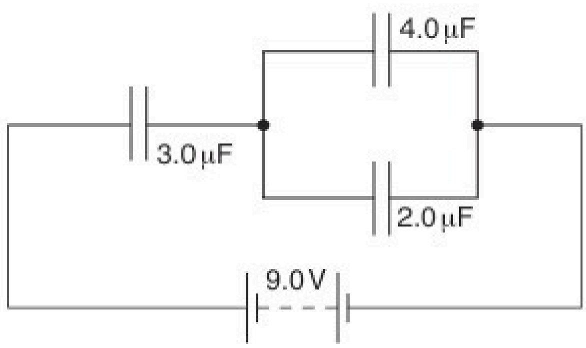


Fig. 7.1

Determine

(i) the combined capacitance of the three capacitors,

capacitance = ... μF [1]

(ii) the potential difference across the capacitor of capacitance 3.0 μF,

potential difference = ... V [2]

(iii) the positive charge stored on the capacitor of capacitance 2.0 μF.

charge = ... μC [2]

[Total: 8]

18. M/J/17/42/Q7

A capacitor consists of two parallel metal plates, separated by an insulator, as shown in Fig. 7.1.

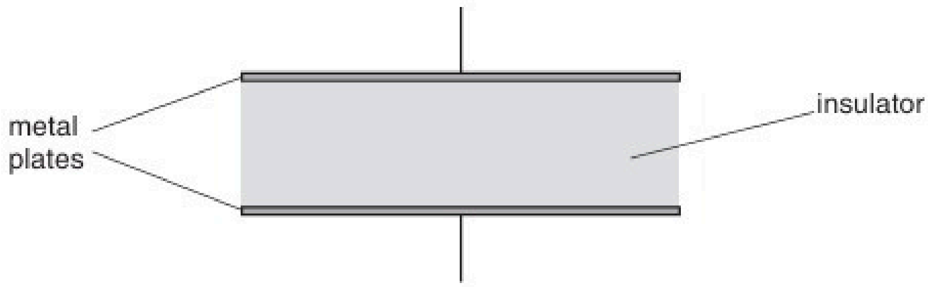


Fig. 7.1

(a) Suggest why, when the capacitor is connected across the terminals of a battery, the capacitor stores energy, not charge.

....

....

....[2]

(b) Define the *capacitance* of the capacitor.

....

....

....[2]

(c) The capacitor is charged so that the potential difference between its plates is V_0.
The capacitor is then connected across a resistor for a short time. It is then disconnected.
The energy stored in the capacitor is reduced to $\frac{1}{16}$ of its initial value.

Determine, in terms of V_0, the potential difference across the capacitor.

potential difference = ...[2]

[Total: 6]

19. O/N/17/41/Q6

Two capacitors P and Q, each of capacitance C, are connected in series with a battery of e.m.f. 9.0 V, as shown in Fig. 6.1.

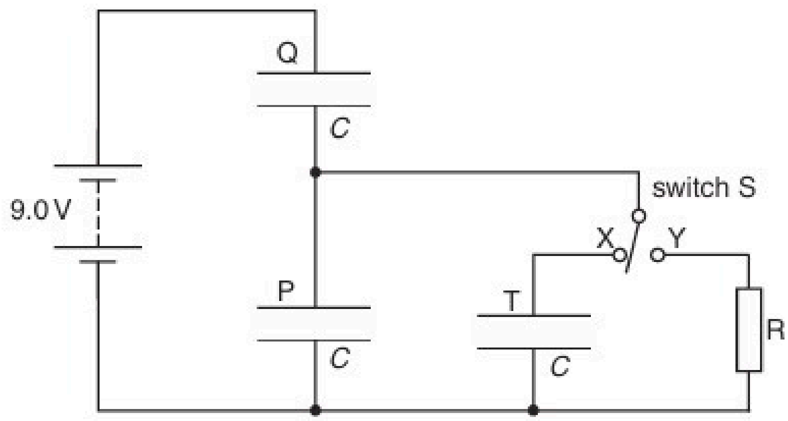


Fig. 6.1

A switch S is used to connect either a third capacitor T, also of capacitance C, or a resistor R, in parallel with capacitor P.

(a) Switch S is in position X.

Calculate

(i) the combined capacitance, in terms of C, of the three capacitors,

capacitance = ... [2]

(ii) the potential difference across capacitor Q. Explain your working.

potential difference = ... V [2]

(b) Switch S is now moved to position Y.
State what happens to the potential difference across capacitor P and across capacitor Q.

capacitor P: ....

....

....

capacitor Q: ....

....

....
[4]

[Total: 8]

20. M/J/18/41/Q7

(a) Explain what is meant by the *capacitance* of a parallel plate capacitor.

....

....

....

....[3]

(b) A parallel plate capacitor C is connected into the circuit shown in Fig. 7.1.

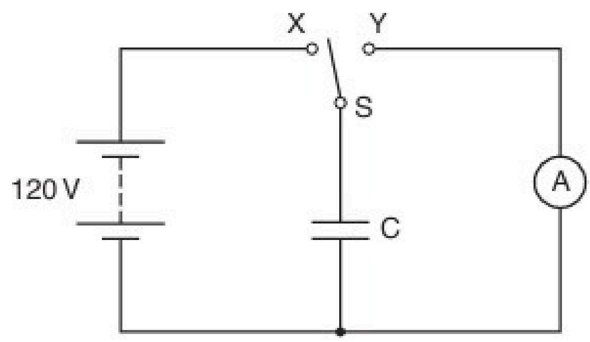


Fig. 7.1

When switch S is at position X, the battery of electromotive force 120 V and negligible internal resistance is connected to capacitor C.
When switch S is at position Y, the capacitor C is discharged through the sensitive ammeter.

The switch vibrates so that it is first in position X, then moves to position Y and then back to position X fifty times each second.

The current recorded on the ammeter is 4.5 μA.

Determine

(i) the charge, in coulomb, passing through the ammeter in 1.0 s,

charge = ... C [1]

(ii) the charge on one plate of the capacitor, each time that it is charged,

charge = ... C [1]

(iii) the capacitance of capacitor C.

capacitance = ... F [2]

(c) A second capacitor, having a capacitance equal to that of capacitor C, is now placed in series with C.

Suggest and explain the effect on the current recorded on the ammeter.

....

....

....[2]

[Total: 9]

21. M/J/18/42/Q6

(a) Explain what is meant by the *capacitance* of a parallel plate capacitor.

....

....

....

....[3]

(b) Three parallel plate capacitors each have a capacitance of 6.0 μF.

Draw circuit diagrams, one in each case, to show how the capacitors may be connected together to give a combined capacitance of

(i) 9.0 μF,

[1]

(ii) 4.0 μF.

[1]

(c) Two capacitors of capacitances 3.0 μF and 2.0 μF are connected in series with a battery of electromotive force (e.m.f.) 8.0 V, as shown in Fig. 6.1.

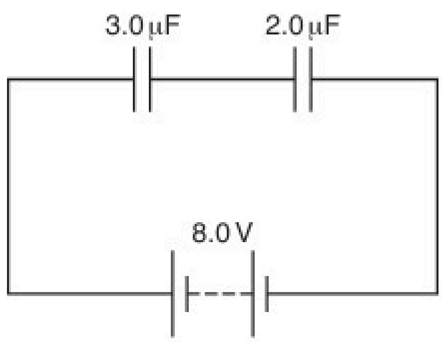


Fig. 6.1

(i) Calculate the combined capacitance of the capacitors.

capacitance = ... μF [1]

(ii) Use your answer in **(i)** to determine, for the capacitor of capacitance 3.0 μF,

1. the charge on one plate of the capacitor,

charge = ... μC

2. the energy stored in the capacitor.

energy = ... J
[4]

[Total: 10]

22. M/J/19/41/Q6

(a) State **two** different functions of capacitors in electrical circuits.

1. ....

....

2. ....

.... [2]

(b) Three uncharged capacitors of capacitances C_1, C_2 and C_3 are connected in series with a battery of electromotive force (e.m.f.) E and a switch, as shown in Fig. 6.1.

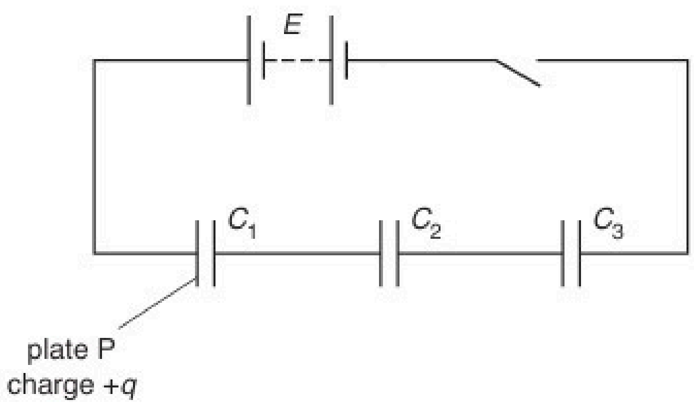


Fig. 6.1

When the switch is closed, there is a charge $+q$ on plate P of the capacitor of capacitance C_1.

Show that the combined capacitance C of the three capacitors is given by the expression

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}.$$

[3]

(c) A student has available four capacitors, each of capacitance 20 μF.

Draw circuit diagrams, one in each case, to show how the student may connect some or all of the capacitors to produce a combined capacitance of:

(i) 60 μF

[1]

(ii) 15 μF.

[1]

[Total: 7]

23. O/N/20/41/Q6

(a) (i) Define the *capacitance* of a parallel plate capacitor.

....

....

.... [2]

(ii) State **three** functions of capacitors in electrical circuits.

1. ....

2. ....

3. ....

[3]

(b) A student has available four capacitors, each of capacitance 24 μF.

The capacitors are connected as shown in Fig. 6.1.

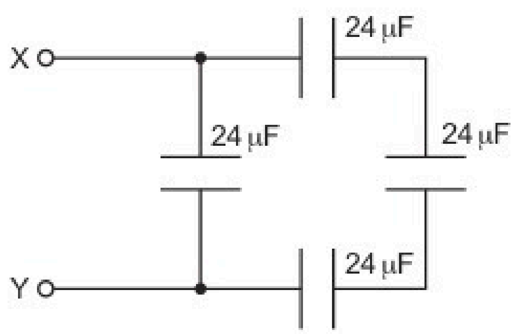


Fig. 6.1

Calculate the combined capacitance between the terminals X and Y.

capacitance = ... μF [2]

[Total: 7]

24. O/N/20/42/Q6

(a) **(i)** Define the *capacitance* of a parallel plate capacitor.

....

....

.... [2]

(ii) State **three** functions of capacitors in electrical circuits.

1. ....

2. ....

3. ....

[3]

(b) A student has available **three** capacitors, each of capacitance 12 μF.

Draw diagrams, one in each case, to show how the student connects the capacitors to give a combined capacitance between the terminals of:

(i) 18 μF

[1]

(ii) 8 μF.

[1]

[Total: 7]

25. M/J/21/41/Q7

(a) State what is meant by the *capacitance* of a parallel plate capacitor.

....

....

.... [2]

(b) A capacitor of capacitance C is connected into the circuit shown in Fig. 7.1.

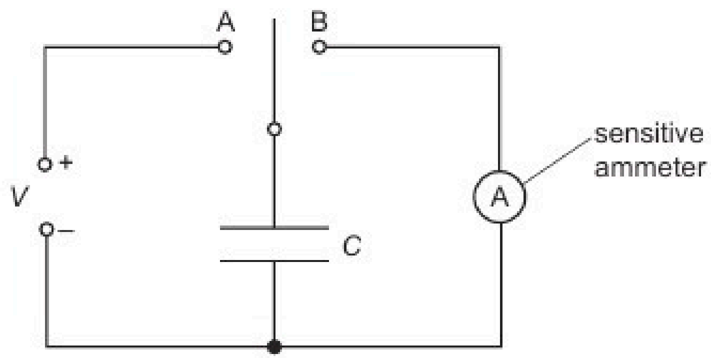


Fig. 7.1

When the two-way switch is in position A, the capacitor is charged so that the potential difference across it is V.
The switch moves to position B and the capacitor fully discharges through the sensitive ammeter.

The switch moves repeatedly between A and B so that the capacitor charges and then discharges with frequency f.

(i) Show that the average current I in the ammeter is given by the expression

$$I = fCV.$$

[2]

(ii) For a potential difference V of 150 V and a frequency f of 60 Hz, the average current in the ammeter is 4.8 μA.

Calculate the capacitance, in pF, of the capacitor.

capacitance = ... pF [2]

(c) A second capacitor, having the same capacitance as the capacitor in **(b)**, is connected into the circuit of Fig. 7.1. The two capacitors are connected in series.

State and explain the new reading on the ammeter.

new reading = ... μA

....

....

.... [3]

[Total: 9]

26. M/J/21/42/Q6

(a) Two flat metal plates are held a small distance apart by means of insulating pads, as shown in Fig. 6.1.

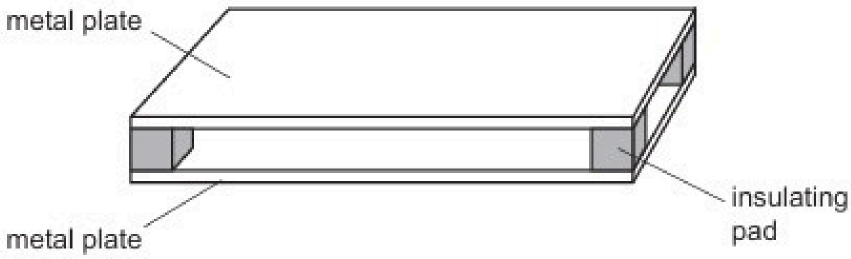


Fig. 6.1

Explain how the plates could act as a capacitor.

....

....

.... [2]

(b) The arrangement in Fig. 6.1 has capacitance *C*.
The arrangement is connected into the circuit of Fig. 6.2.

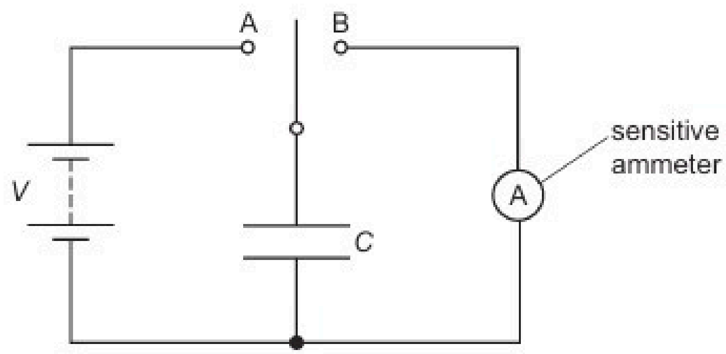


Fig. 6.2

When the two-way switch is moved to position A, the capacitor is charged so that the potential difference across it is *V*. When the switch moves to position B, the capacitor fully discharges through the sensitive ammeter.

The switch moves repeatedly between A and B so that the capacitor charges and then discharges with frequency *f*.

(i) Show that the average current I in the ammeter is given by

$$I = CVf.$$

[2]

(ii) For a potential difference V of 180 V and a frequency f of switching of 50 Hz, the average current I in the ammeter is 2.5 μA.

Calculate the capacitance, in pF, of the parallel plates.

capacitance = ... pF [2]

(c) A second capacitor is connected into the circuit of Fig. 6.2.
The two capacitors are connected in parallel.

State and explain the change, if any, in the average current in the ammeter.

....

....

.... [2]

[Total: 8]

27. O/N/21/41/Q6

(a) A capacitor consists of two parallel metal plates, separated by air, at a variable distance x apart, as shown in Fig. 6.1. The capacitance C is inversely proportional to x.

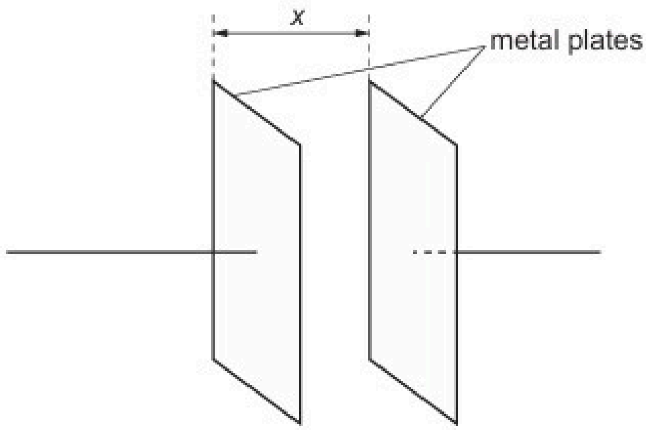


Fig. 6.1

The capacitor is charged by a supply so that there is a potential difference (p.d.) V between the plates.

State expressions, in terms of C and V, for the charge Q on one of the plates and for the energy E stored in the capacitor.

Q = ... E = ... [1]

(b) The charged capacitor in **(a)** is now disconnected from the supply. The plates of the capacitor are initially separated by distance L. They are then moved closer together by a distance D, as shown in Fig. 6.2.

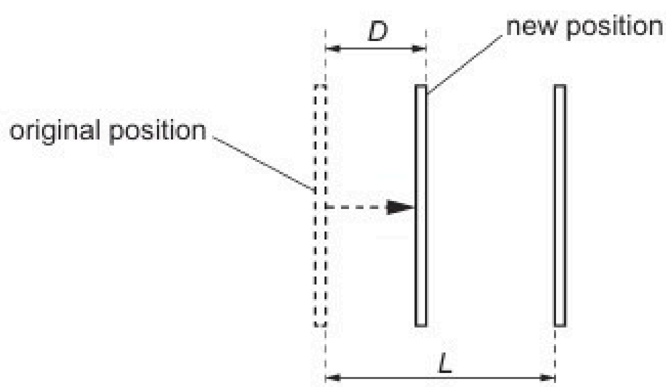


Fig. 6.2

State expressions, in terms of C, V, L and D, for:

(i) the new capacitance C_N

C_N = ... [1]

(ii) the new charge Q_N on one of the plates

Q_N = ... [1]

(iii) the new p.d. V_N between the plates.

V_N = ... [1]

(c) Explain whether reducing the separation of the plates in **(b)** results in an increase or decrease in the energy stored in the capacitor.

....

....

.... [1]

[Total: 5]

28. F/M/22/42/Q5

The variation with potential difference V of the charge Q on one of the plates of a capacitor is shown in Fig. 5.1.

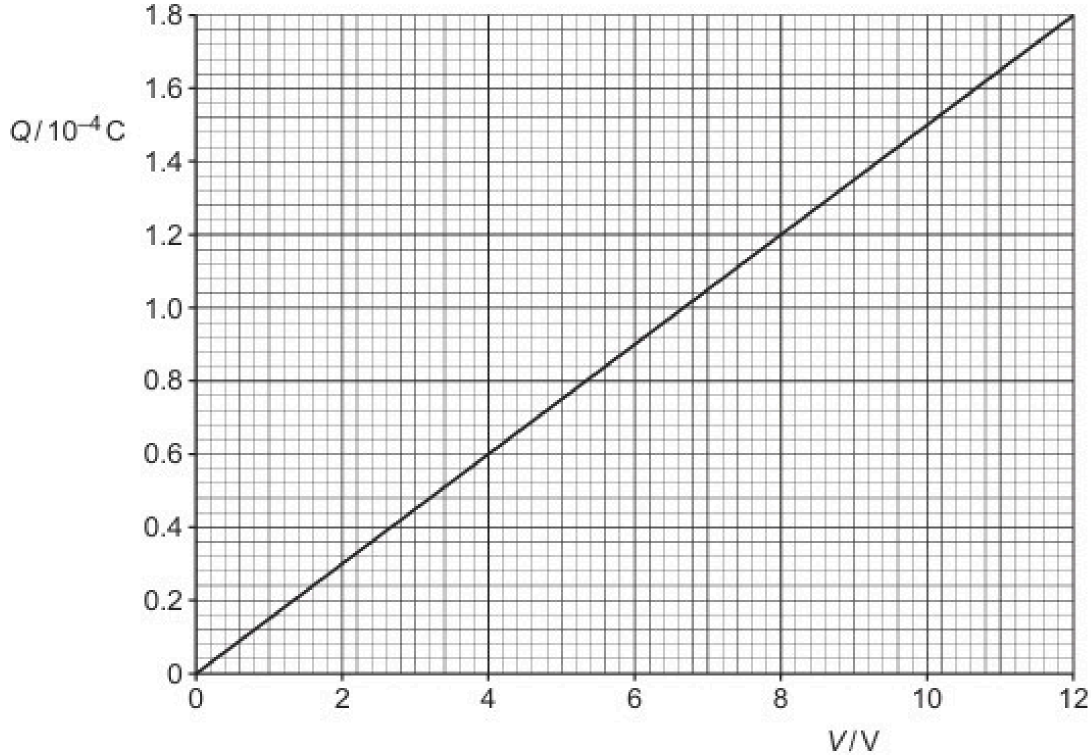


Fig. 5.1

The capacitor is connected to an 8.0V power supply and two resistors R and S as shown in Fig. 5.2.

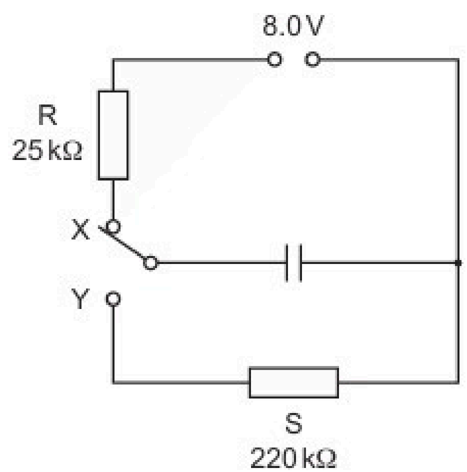


Fig. 5.2

The resistance of R is 25kΩ and the resistance of S is 220kΩ.

The switch can be in either position X or position Y.

(a) The switch is in position X so that the capacitor is fully charged.

Calculate the energy *E* stored in the capacitor.

E = ... J [2]

(b) The switch is now moved to position Y.

(i) Show that the time constant of the discharge circuit is 3.3 s.

[2]

(ii) The fully charged capacitor in **(a)** stores energy *E*.

Determine the time *t* taken for the stored energy to decrease from *E* to *E*/9.

t = ... s [4]

(c) A second identical capacitor is connected in parallel with the first capacitor.

State and explain the change, if any, to the time constant of the discharge circuit.

....

....

.... [2]

[Total: 10]

29. M/J/22/42/Q5

(a) Define the capacitance of a parallel plate capacitor.

....

....

.... [2]

(b) Two capacitors, of capacitances C_1 and C_2, are connected in parallel to a power supply of electromotive force (e.m.f.) E, as shown in Fig. 5.1.

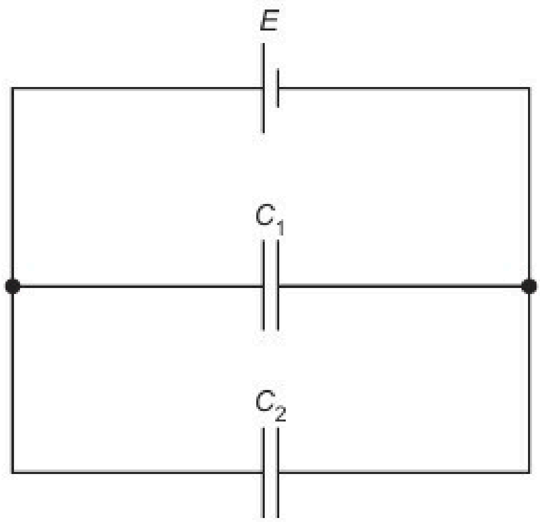


Fig. 5.1

Show that the combined capacitance C_T of the two capacitors is given by

$$C_T = C_1 + C_2.$$

Explain your reasoning. You may draw on Fig. 5.1 if you wish.

[3]

(c) Two capacitors of capacitances 22 μF and 47 μF, and a resistor of resistance 2.7 MΩ, are connected into the circuit of Fig. 5.2.

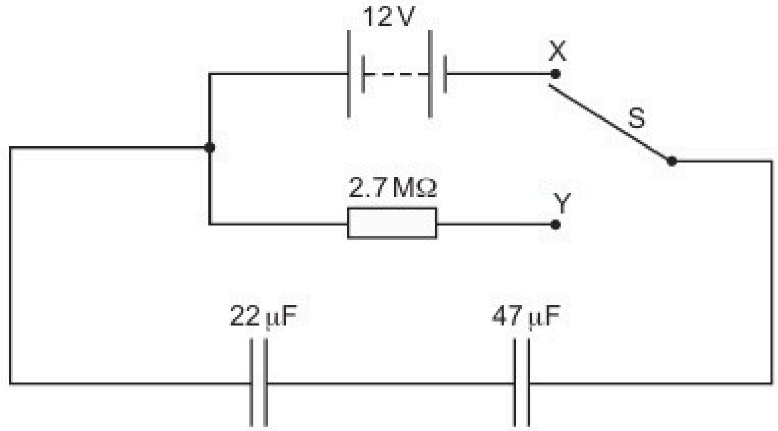


Fig. 5.2

The battery has an e.m.f. of 12 V.

(i) Show that the combined capacitance of the two capacitors is 15 μF.

[1]

(ii) The two-way switch S is initially at position X, so that the capacitors are fully charged.

Use the information in **(c)(i)** to calculate the total energy stored in the two capacitors.

total energy = ... J [2]

(iii) The two-way switch is now moved to position Y.

Determine the time taken for the potential difference (p.d.) across the 22 μF capacitor to become 6.0 V.

time = ... s [3]

30. O/N/22/41/Q5

A capacitor of capacitance 470 μF is connected to a battery of electromotive force (e.m.f.) 24 V in the circuit of Fig. 5.1.

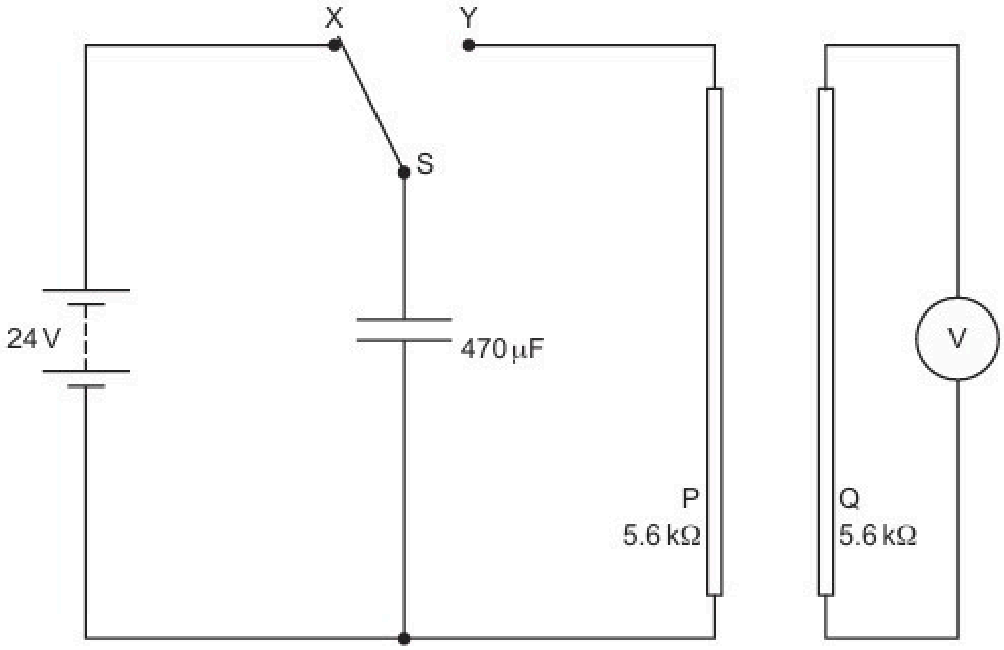


Fig. 5.1

The two-way switch S is initially at position X.

P and Q are identical long straight wires, each with a resistance of 5.6 kΩ. These wires are placed near to, and parallel to, each other. Wire Q is connected to a voltmeter.

At time $t = 0$, switch S is moved to position Y so that the capacitor discharges through wire P.

(a) **(i)** Calculate the charge Q_0 on the capacitor at time $t = 0$.

Q_0 = ... C [2]

(ii) Calculate the current I_0 in wire P at time $t = 0$.

I_0 = ... A [1]

(iii) Calculate the time constant τ of the discharge circuit.

τ = ... s [2]

(iv) On Fig. 5.2, sketch a line to show the variation with t of the current I in wire P as the capacitor discharges.

Fig. 5.2

[2]

(b) (i) Explain why there is an induced e.m.f. across wire Q during the discharge of the capacitor.

....

....

....

.... [3]

(ii) On Fig. 5.3, sketch a line to suggest the variation with t of the voltmeter reading V.

Fig. 5.3

[1]

[Total: 11]

31. O/N/22/42/Q6

A capacitor of capacitance C and a resistor of resistance R are connected as shown in Fig. 6.1.

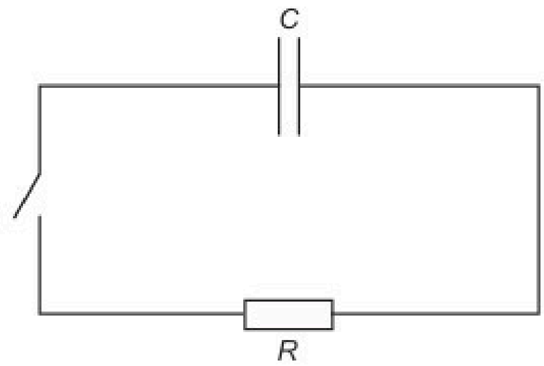


Fig. 6.1

Initially, the capacitor is charged and the switch is open.

The switch is closed at time $t = 0$.

Fig. 6.2 and Fig. 6.3 show, respectively, the variations with t of the charge Q on the capacitor and the potential difference (p.d.) V across the resistor.

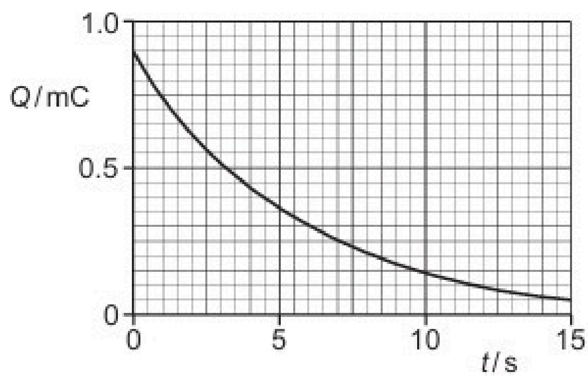


Fig. 6.2

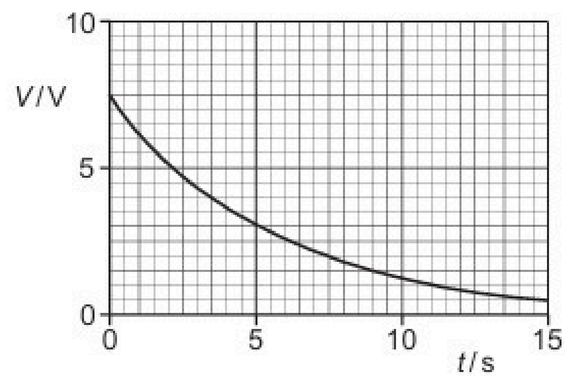


Fig. 6.3

(a) Explain the shape of the line in Fig. 6.3 representing the variation of V with t.

....

....

....

....

.... [3]

(b) Use Fig. 6.2 to show that the time constant of the circuit in Fig. 6.1 is 5.5 s.

[3]

(c) Use Fig. 6.2, Fig. 6.3 and the information in **(b)** to determine:

(i) capacitance C, in μF

C = ... μF [2]

(ii) resistance R, in kΩ.

R = ... kΩ [2]

[Total: 10]

32. *F/M/23/42/Q5*

A capacitor, a battery of electromotive force (e.m.f.) 12 V, a resistor R and a two-way switch are connected in the circuit shown in Fig. 5.1.

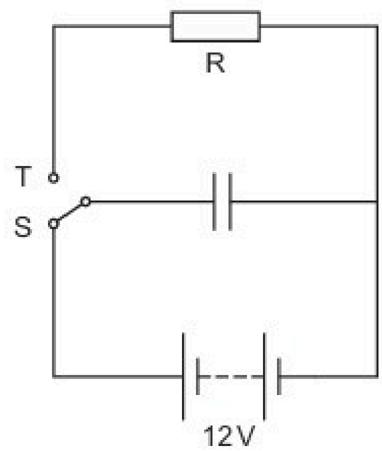


Fig. 5.1

The switch is initially in position S. When the capacitor is fully charged, the switch is moved to position T so that the capacitor discharges. At time t after the switch is moved the charge on the capacitor is Q.

The variation with t of $\ln(Q/\mu C)$ is shown in Fig. 5.2.

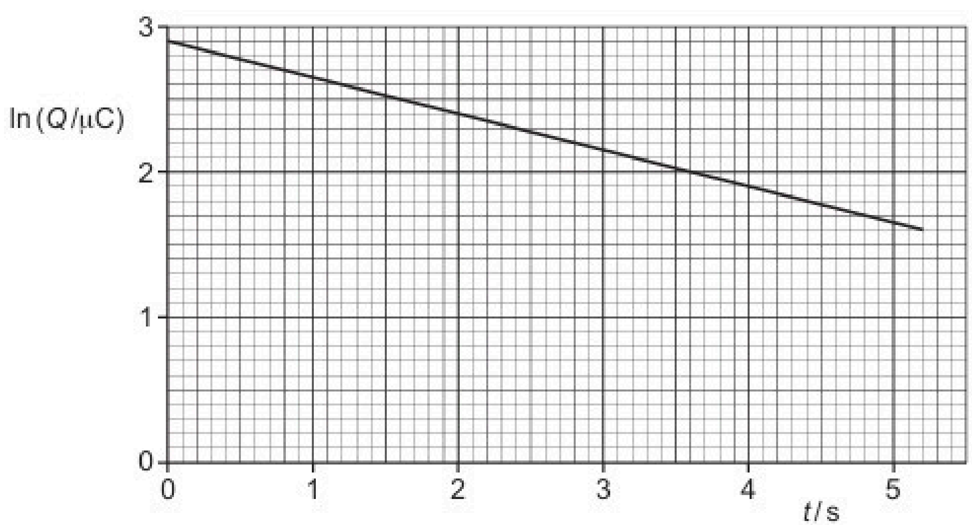


Fig. 5.2

(a) Show that the capacitance of the capacitor is $1.5\,\mu F$.

[3]

(b) Determine the resistance of R.

resistance = ... Ω [3]

(c) Calculate the energy stored in the capacitor at time $t = 0$.

energy = ... J [2]

(d) A second identical resistor is now connected in parallel with R.

The switch is initially in position S. When the capacitor is fully charged, the switch is moved to position T so that the capacitor discharges. At time t after the switch is moved the charge on the capacitor is Q.

On Fig. 5.2, sketch a line to show the variation of $\ln(Q/\mu C)$ with t between time $t = 0$ and time $t = 5.0\,s$. [2]

[Total: 10]

33. M/J/23/42/Q5

Two capacitors A and B are connected into the circuit shown in Fig. 5.1.

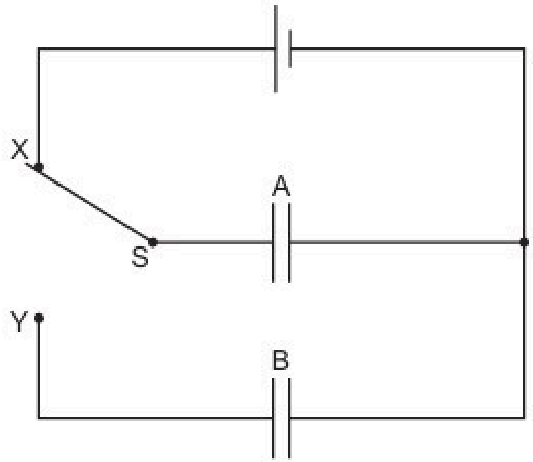


Fig. 5.1

Capacitor A has capacitance *C* and capacitor B has capacitance 3*C*.
The electromotive force (e.m.f.) of the cell is *V*.
The two-way switch S is initially at position X, and capacitor B is initially uncharged.

(a) State, in terms of *V* and *C*, expressions for:

(i) the initial charge Q_A on the plates of capacitor A

Q_A = ... [1]

(ii) the initial energy E_A stored in capacitor A.

E_A = ... [1]

(b) The two-way switch S is now moved to position Y.

(i) State and explain what happens to the charge that was initially on the plates of capacitor A.

....

....

.... [2]

(ii) Show that the final potential difference (p.d.) V_B across capacitor B is given by

$$V_B = \frac{V}{4}.$$

Explain your reasoning.

[3]

(iii) Determine an expression, in terms of V and C, for the decrease ΔE in the total energy that is stored in the capacitors as a result of the change of the position of the switch.

ΔE = ... [2]

[Total: 9]

34. O/N/23/42/Q6

A capacitor C is charged so that the potential difference (p.d.) V across its terminals is 8.0 V.
The capacitor is connected into the circuit of Fig. 6.1.

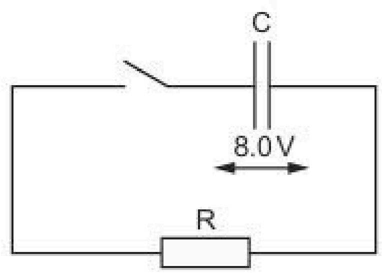


Fig. 6.1

The switch is initially open. The switch is closed at time t = 0.

(a) Fig. 6.2 shows the variation of V with the charge Q on the plates of capacitor C as the capacitor discharges.

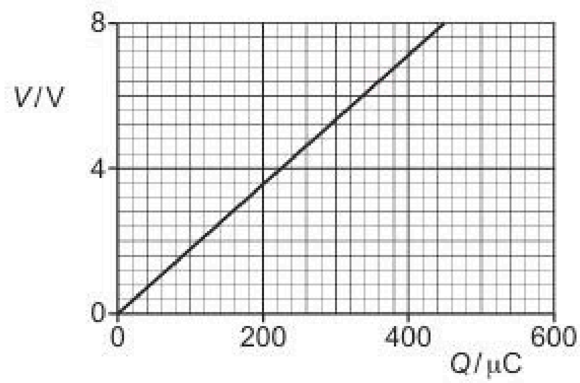


Fig. 6.2

(i) Show that the energy stored in capacitor C at time t = 0 is 1.8 mJ.

[2]

(ii) Determine the capacitance of capacitor C. Give a unit with your answer.

capacitance = ... unit .... [2]

(b) Fig. 6.3 shows the variation with t of $-\ln\left(\frac{V}{8.0\,\text{V}}\right)$.

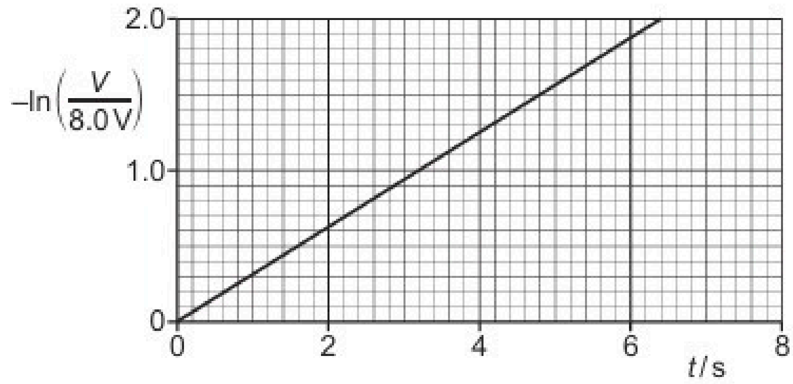


Fig. 6.3

(i) Show that, when t is equal to one time constant, the value of $-\ln\left(\frac{V}{8.0\,\text{V}}\right)$ is equal to 1.0.

[2]

(ii) Determine the time constant τ of the circuit in Fig. 6.1.

τ = ... s [1]

(iii) Calculate the resistance of resistor R.

resistance = ... Ω [2]

[Total: 9]

ANSWERS

1.

(a) e.g. separate charges, store energy, smoothing circuit. etc. B1
(*allow 'stores charge'*)

(b) **(i)** charge = current × time B1

(ii) area is 21.2 cm^2 (*allow* ± 0.5 *cm^2*) C2
(*allow 1 mark if outside* ± 0.5 *cm^2 but within* ± 1.0 *cm^2*)
1.0 cm^2 represents ($0.125 \times 10^{-3} \times 1.25$ =) 156 μC C1
charge = 3300 μC A1

(iii) capacitance = Q/V C1
$= (3300 \times 10^{-6}) / 15$
= 220 μF A1

(c) *either* energy = $\frac{1}{2}CV^2$ or energy = $\frac{1}{2}QV$ and $C = Q/V$ C1
$\frac{1}{2} \times C \times 15^2 = 2 \times \frac{1}{2} \times C \times V^2$ C1
V = 10.6 V A1

2.

(a) at t = 1.0 s, V = 2.5 V C1
energy = $\frac{1}{2}CV^2$ C1
$0.13 = \frac{1}{2} \times C \times (8.0^2 - 2.5^2)$ M1
C = 4500 μF A0

(b) use of two capacitors in series in all branches of combination M1
connected into correct parallel arrangement A1

3.

(a) **(i)** ratio of charge (on body) and its potential B1
(do not allow reference to plates of a capacitor)

(ii) (potential at surface of sphere =) $V = Q / 4\pi\varepsilon_0 r$ M1
$C = Q / V = 4\pi\varepsilon_0 r$ A0

(b) **(i)** $C = 4 \times \pi \times 8.85 \times 10^{-12} \times 0.36$
$= 4.0 \times 10^{-11}$ F *(allow 1 s.f.)* A1

(ii) $Q = CV$
$= 4.0 \times 10^{-11} \times 7.0 \times 10^5$
$= 2.8 \times 10^{-5}$ C A1

(c) plastic is an insulator / not a conductor / has no free electrons B1
charges do not move (on an insulator) B1
either so no single value for the potential
or charge cannot be considered to be at centre B1

(d) *either* energy = $\frac{1}{2}CV^2$ *or* energy = $\frac{1}{2}QV$ and $C = Q/V$ C1
energy = $\frac{1}{2} \times 4 \times 10^{-11} \times \{(7.0 \times 10^5)^2 - (2.5 \times 10^5)^2\}$ C1
= 8.6 J A1

4.

(a) charge / potential ………(*ratio must be clear*) B1

(b) potential (at surface of sphere) = $Q / 4\pi\varepsilon_0 R$ M1
$C = Q / V = 4\pi\varepsilon_0 R$ A0

(c) **(i)** $C = 4\pi \times 8.85 \times 10^{-12} \times 0.63$ C1
$= 7.0 \times 10^{-11}$ A1
farad / F B1

(ii) energy = $\frac{1}{2}CV^2$ C1
$0.25 \times \frac{1}{2}C \times (1.2 \times 10^6)^2 = \frac{1}{2}CV^2$ C1
$V = 6.0 \times 10^5$ V A1
(*use of 0.75 rather than 0.25, allow max 2 marks*)

5.

(a) e.g. 'storage of charge' / storage of energy
blocking of direct current
producing of electrical oscillations
smoothing
(any two, 1 mark each) B2

(b) **(i)** capacitance of parallel combination = 60 μF C1
total capacitance = 20 μF A1

(ii) p.d. across parallel combination = $\frac{1}{2} \times$ p.d. across single capacitor C1
maximum is 9V A1

(c) *either* energy = $\frac{1}{2}CV^2$ *or* energy = $\frac{1}{2}QV$ and $Q = CV$ C1
energy = $\frac{1}{2} \times 4700 \times 10^{-6} \times (18^2 - 12^2)$ C1
= 0.42 J A1

6.

(a)	charge / potential (difference) (*ratio must be clear)*	B1
(b) (i)	$V = Q / 4\pi\varepsilon_0 r$	B1
(ii)	$C = Q / V = 4\pi\varepsilon_0 r$ and <u>$4\pi\varepsilon_0$ is constant</u>	M1
	so $C \propto r$	A0
(c) (i)	$r = C / 4\pi\varepsilon_0 r$	C1
	$r = (6.8 \times 10^{-12}) / (4\pi \times 8.85 \times 10^{-12})$	C1
	$= 6.1 \times 10^{-2}$ m	A1
(ii)	$Q = CV = 6.8 \times 10^{-12} \times 220$	
	$= 1.5 \times 10^{-9}$ C	A1
(d) (i)	$V = Q/C = (1.5 \times 10^{-9}) / (18 \times 10^{-12})$	
	$= 83$ V	A1
(ii)	*either* energy $= ½CV^2$	C1
	$\Delta E = ½ \times 6.8 \times 10^{-12} \times 220^2 - ½ \times 18 \times 10^{-12} \times 83^2$	C1
	$= 1.65 \times 10^{-7} - 6.2 \times 10^{-8}$	
	$= 1.03 \times 10^{-7}$ J	A1
	or energy $= ½QV$	(C1)
	$\Delta E = ½ \times 1.5 \times 10^{-9} \times 220 - ½ \times 1.5 \times 10^{-9} \times 83$	(C1)
	$= 1.03 \times 10^{-7}$ J	(A1)

7.

(a) (i)	work done moving unit positive charge	M1
	from infinity <u>to the point</u>	A1
(ii)	charge / potential (difference) (*ratio must be clear*)	B1
(b) (i)	capacitance $= (2.7 \times 10^{-6}) / (150 \times 10^3)$	C1
	(*allow any appropriate values*)	
	capacitance $= 1.8 \times 10^{-11}$ (*allow 1.8 ±0.05)*	A1
(ii)	*either* energy $= ½CV^2$ *or* energy $= ½QV$ <u>*and*</u> $Q = CV$	C1
	energy $= ½ \times 1.8 \times 10^{-11} \times (150 \times 10^3)^2$ *or* $½ \times 2.7 \times 10^{-6} \times 150 \times 10^3$	
	$= 0.20$ J	A1
(c)	*either* since energy $\propto V^2$, capacitor has $(½)^2$ of its energy left	
	or full formula treatment	C1
	energy lost = 0.15 J	A1

8.

(a) charges on plates are equal and opposite — M1
so no resultant charge — A1
energy stored because there is charge separation — B1

(b) **(i)** capacitance $= Q / V$ — C1
$= (18 \times 10^{-3}) / 10$
$= 1800\ \mu F$ — A1

(ii) use of area under graph *or* energy $= \frac{1}{2}CV^2$ — C1
energy $= 2.5 \times 15.7 \times 10^{-3}$ *or* energy $= \frac{1}{2} \times 1800 \times 10^{-6} \times (10^2 - 7.5^2)$
$= 39\,mJ$ — A1

(c) combined capacitance of Y & Z $= 20\,\mu F$ *or* total capacitance $= 6.67\,\mu F$ — C1
p.d. across capacitor X $= 8\,V$ *or* p.d. across combination $= 12\,V$ — C1
charge $= 10 \times 10^{-6} \times 8$ *or* $6.67 \times 10^{-6} \times 12$
$= 80\,\mu C$ — A1

9.

(a) e.g. storing energy
separating charge
blocking d.c.
producing electrical oscillations
tuning circuits
smoothing
preventing sparks
timing circuits
(any two sensible suggestions, 1 each, max 2) — B2

(b) **(i)** $-Q$ (induced) on opposite plate of C_1 — B1
by <u>charge conservation</u>, charges are $-Q$, $+Q$, $-Q$, $+Q$, $-Q$ — B1

(ii) total p.d. $V = V_1 + V_2 + V_3$ — B1
$Q/C = Q/C_1 + Q/C_2 + Q/C_3$ — B1
$1/C = 1/C_1 + 1/C_2 + 1/C_3$ — A0

(c) **(i)** energy $= \frac{1}{2}CV^2$ or energy $= \frac{1}{2} QV$ and $C = Q/V$ — C1
$= \frac{1}{2} \times 12 \times 10^{-6} \times 9.0^2$
$= 4.9 \times 10^{-4}$ J — A1

(ii) energy dissipated in (resistance of) wire/as a spark — B1

10.

(a) **(i)** ratio of charge and potential (difference) / voltage
(*ratio must be clear*) B1

(ii) capacitor has equal magnitudes of (+)ve and (-)ve charge B1
<u>total</u> charge on capacitor is zero (so does not store charge) B1
(+)ve and (-)ve charges to be separated M1
work done to achieve this so stores energy A1

(b) **(i)** capacitance of Y and Z together is 24 μF C1
$1/C = 1/24 + 1/12$
$C = 8.0$ μF (*allow* 1 s.f.) A1

(ii) some discussion as to why all charge of one sign on one plate of X B1
$Q = (CV =)\ \underline{8.0 \times 10^{-6}} \times 9.0$ M1
$= 72\,\mu C$ A0

(iii) **1.** $V = (72 \times 10^{-6}) / (12 \times 10^{-6})$
$= 6.0\,V$ (*allow* 1 s.f.) (allow 72/12) A1

2. *either* $Q = 12 \times 10^{-6} \times 3.0$ *or* charge is shared between Y and Z C1
charge = $36\,\mu C$ A1
Must have correct voltage in ***(iii)****1 if just quote of 36μC in* ***(iii)****2.*

11.

(a) **(i)** $V = q / 4\pi\varepsilon_0 R$ B1

(ii) (capacitance is) ratio of charge and potential or q/V M1
$C = q/V = 4\pi\varepsilon_0 R$ A0

(b) **(i)** $C = 4\pi \times 8.85 \times 10^{-12} \times 0.45$ C1
$= 50$ pF A1

(ii) either energy = $\frac{1}{2} CV^2$ or energy = $\frac{1}{2} QV$ and $Q = CV$ C1
energy of spark = $\frac{1}{2} \times 50 \times 10^{-12} \{(9.0 \times 10^5)^2 - (3.6 \times 10^5)^2\}$ C1
$= 17$ J A1

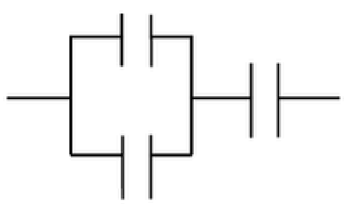
12.

(a) e.g. store energy (do not allow 'store charge')
in smoothing circuits
blocking d.c.
in oscillators
any sensible suggestions, one each, max. 2 B2

(b) **(i)** potential across each capacitor is the same <u>and</u> $Q = CV$ B1

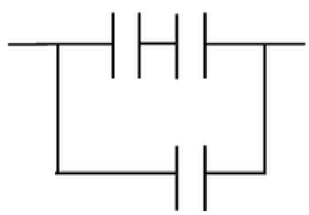
(ii) total charge $Q = Q_1 + Q_2 + Q_3$ M1
$CV = C_1V + C_2V + C_3V$ M1
(*allow* $Q = CV$ *here* or in **(i)**)
so $C = C_1 + C_2 + C_3$ A0

(c) **(i)**



A1

(ii)



A1

13.

(a) **(i)** energy $= EQ$ C1
$= 9.0 \times 22 \times 10^{-3}$
$= 0.20$ J A1

(ii) **1.** $C = Q / V$
$V = (22 \times 10^{-3})/(4700 \times 10^{-6})$ C1
$= 4.7$ V A1

2. *either* $E = ½CV^2$ C1
$= ½ \times 4700 \times 10^{-6} \times 4.7^2$
$= 5.1 \times 10^{-2}$ J A1

or $E = ½QV$ (C1)
$= ½ \times 22 \times 10^{-3} \times 4.7$
$= 5.1 \times 10^{-2}$ J (A1)

or $E = ½Q^2/C$ (C1)
$= ½ \times (22 \times 10^{-3})^2/4700 \times 10^{-6}$
$= 5.1 \times 10^{-2}$ J (A1)

(b) energy lost (as thermal energy) in resistance/wires/battery/resistor — B1
*(award only if answer in **(a)(i)** > answer in **(a)(ii)2**)*

14.

(a) for the two capacitors in parallel, capacitance = 96 μF — C1
for complete arrangement, $1/C_T = 1/96 + 1/48$
$C_T = 32\,\mu$F — A1

(b) p.d. across parallel combination is one half p.d. across single capacitor — C1
total p.d. = 9 V — A1

15.

(a) (capacitance =) charge / potential (difference) — B1

(b) $V = V_1 + V_2 + V_3$ — B1
either $Q/C = Q/C_1 + Q/C_2 + Q/C_3$ *or* $V/Q = V_1/Q + V_2/Q + V_3/Q$
and so $1/C = 1/C_1 + 1/C_2 + 1/C_3$ — B1

(c) **(i)** **1.** $1/C_T = (1/200) + (1/600)$
$C_T = 150\,\mu$F — A1

2. $Q = CV$
$= 150 \times 10^{-6} \times 12$ *or* $600 \times 10^{-6} \times 3.0$ *or* $200 \times 10^{-6} \times 9.0$
$= 1.8 \times 10^{-3}$ C — A1

3. $V = Q/C = 1.8 \times 10^{-3}/600 \times 10^{-6}$ *or* $V = [200/(200 + 600)] \times 12$
$= 3(.0)$ V — A1

(ii) energy = $\frac{1}{2}CV^2$ or energy = $\frac{1}{2}QV$ and $C = Q/V$ — C1
$\frac{1}{2} \times C \times 3^2 = 2 \times \frac{1}{2} \times C \times V^2$ — C1
$V = 2.1$ V — A1

16.

(a) e.g. storing energy
blocking d.c.
in oscillator circuits
in tuning circuits
in timing circuits

any two — B2

(b) (i)	$1/6 + 1/C + 1/C = 1/4$	C1
	$C = 24\,\mu F$	A1
(ii)	$Q = CV$ $= 4.0 \times 10^{-6} \times 12$	C1
	$= 48\,\mu C$	A1
(iii)	**1.** $48\,\mu C$	A1
	2. $24\,\mu C$	A1

17.

(a) (i)	charge / potential (difference) or charge per (unit) potential (difference)	B1
(ii)	($V = Q/4\pi\varepsilon_0 r$ and $C = Q/V$)	
	for sphere, C (= Q/V) = $4\pi\varepsilon_0 r$	C1
	$C = 4\pi \times 8.85 \times 10^{-12} \times 12.5 \times 10^{-2} = 1.4 \times 10^{-11}\,F$	A1
(b) (i)	$1/C_T = 1/3.0 + 1/6.0$	
	$C_T = 2.0\,\mu F$	A1
(ii)	total charge = charge on $3.0\,\mu F$ capacitor = $2.0\,(\mu) \times 9.0 = 18\,(\mu C)$	C1
	potential difference = $Q/C = 18\,(\mu)C / 3.0\,(\mu)F = 6.0\,V$	A1
	or	
	argument based on equal charges:	
	$3.0 \times V = 6.0 \times (9.0 - V)$	(C1)
	$V = 6.0\,V$	(A1)
(iii)	potential difference (= $9.0 - 6.0$) = $3.0\,V$	C1
	charge (= $3.0 \times 2.0\,(\mu)$) = $6.0\,\mu C$	A1

18.

7(a)	equal and opposite charges on the plates so no resultant charge	**B1**
	+ve and –ve charges separated so energy stored	**B1**
7(b)	charge / potential difference	**M1**
	reference to charge on one plate and p.d. between plates	**A1**
7(c)	energy = ½ CV^2 **or** energy = ½ QV **and** $C = Q / V$	**C1**
	$(1 / 16) \times ½\, CV_0^2 = ½\, CV^2$ $V = ¼\, V_0$	**A1**

19.

6(a)(i)	$1 / T = 1 / (2C) + 1 / C$	**C1**
	$T = ⅔C$ **or** $0.67C$	**A1**
6(a)(ii)	same charge on Q as on combination	**B1**
	so p.d. is 6.0 V	**B1**
6(b)	P: p.d. will decrease (from 3.0 V)	**B1**
	to zero	**B1**
	Q: p.d. will increase (from 6.0 V)	**B1**
	to 9.0 V	**B1**

20.

7(a)	(capacitance =) charge / potential	**M1**
	charge is (numerically equal to) charge on one plate	**A1**
	potential is potential difference between plates	**A1**
7(b)(i)	4.5×10^{-6} C	**A1**
7(b)(ii)	9.0×10^{-8} C	**A1**
7(b)(iii)	capacitance = $(9.0 \times 10^{-8}) / 120$	**C1**
	= 7.5×10^{-10} F	**A1**
7(c)	total capacitance is halved	**B1**
	current is halved	**B1**

21.

6(a)	capacitance = charge / potential	**M1**
	charge is (numerically equal to) charge on one plate	**A1**
	potential is potential difference between plates	**A1**
6(b)(i)	two in series, in parallel with the other (correct symbols)	**A1**
6(b)(ii)	two in parallel connected to one in series (correct symbols)	**A1**
6(c)(i)	capacitance = 1.2 μF	**A1**
6(c)(ii)	**1.** $Q = CV$	**C1**
	$= 1.2 \times 8.0$ $= 9.6\ \mu C$	**A1**
	2. $E = ½QV$ **and** $V = Q/C$ **or** $E = ½CV^2$ **and** $V = Q/C$ **or** $E = ½Q^2/C$	**C1**
	$E = ½\,(9.6 \times 10^{-6})^2 / (3.0 \times 10^{-6})$ $= 1.5 \times 10^{-5}$ J	**A1**

22.

6(a)	Any valid two points e.g.: • to store (electrical) energy • smoothing/reduce ripple (on direct voltages/currents) • to block d.c. • timing/time delay (circuits) • in oscillator (circuits) • in tuning (circuits) • to prevent arcing/sparks	**B2**
6(b)	clear indication of equal charge on each capacitor	**B1**
	$E = V_1 + V_2 + V_3$ **and** $V = Q/C$	**M1**
	completion of algebra leading to $1/C = 1/C_1 + 1/C_2 + 1/C_3$	**A1**
6(c)(i)	three capacitors connected in parallel	**B1**
6(c)(ii)	parallel combination of three capacitors connected in series with one capacitor	**B1**

23.

6(a)(i)	charge per unit potential (difference)	**M1**
	charge on one plate <u>and</u> potential difference across the plates	**A1**
6(a)(ii)	any three points from: • smoothing • timing/(time) delay • tuning • oscillator • blocking d.c. • surge protection • temporary power supply	**B3**
6(b)	(capacitors in series have combined capacitance =) 8 μF	**C1**
	capacitance = 8 + 24 = 32 μF	**A1**

24.

6(a)(i)	charge per unit potential (difference)	**M1**
	charge on one plate and potential difference between the plates	**A1**
6(a)(ii)	any three points from: • smoothing • timing/(time) delaying • tuning • oscillator • blocking d.c. • surge protection • temporary power supply	**B3**
6(b)(i)	parallel combination of two in series and a single capacitor	**B1**
6(b)(ii)	one capacitor in series with two in parallel	**B1**

25.

7(a)	charge / potential	**M1**
	charge is on one plate, potential is p.d. between the plates	**A1**
7(b)(i)	$I = Q / t$	**M1**
	charge = CV <u>and</u> time = $1 / f$ leading to $I = fCV$	**A1**
7(b)(ii)	$4.8 \times 10^{-6} = 150 \times 60 \times C$	**C1**
	C = 530 pF	**A1**
7(c)	(total) capacitance is halved	**B1**
	charge (for each cycle/discharge) is halved **or** since f and V are constant, current is proportional to capacitance	**B1**
	current = 2.4 μA	**B1**

26.

6(a)	potential difference applied between the plates	**M1**
	causes charge separation (between the plates) **or** causes energy to be stored (between the plates)	**A1**
6(b)(i)	$I = Q / t$	**M1**
	clear substitution of $Q = CV$ and $f = 1 / t$, leading to $I = fCV$	**A1**
6(b)(ii)	$2.5 \times 10^{-6} = 50 \times C \times 180$	**C1**
	C = 280 pF	**A1**
6(c)	(total) capacitance increases	**B1**
	greater charge (for each cycle/discharge) so greater (average) current **or** V and f are constant so (average) current increases **or** I is (directly) proportional to C so (average) current increases	**B1**

27.

6(a)	$Q = CV$ **and** $E = \frac{1}{2}CV^2$	**B1**
6(b)(i)	$C_N = CL / (L - D)$	**B1**
6(b)(ii)	(charge is unchanged by moving the plates so) $Q_N = CV$	**B1**
6(b)(iii)	$V_N = Q_N / C_N$ $= (CV) / [CL / (L - D)]$ $= V(L - D) / L$	**B1**
6(c)	oppositely charged plates attract, so energy stored decreases	**B1**

28.

5(a)	(energy stored =) area under line or ½ QV $= \frac{1}{2} \times 8.0 \times 1.2 \times 10^{-4}$	**C1**
	$=$ **4.8×10^{-4}** J	**A1**
5(b)(i)	(τ=) RC	**C1**
	(τ=) $220 \times 10^3 \times (1.2 \times 10^{-4}/8.0) =$ **3.3** s	**A1**
5(b)(ii)	$E \propto V^2$	**C1**
	(so time to) $V_o / 3$ $V = V_o e^{-t/RC}$	**C1**
	$\frac{V_o}{3} = V_o e^{-t/3.3}$ $\frac{1}{3} = e^{-t/3.3}$	**C1**
	t = 3.6 s	**A1**
5(c)	(total) capacitance is doubled	**M1**
	time constant is doubled	**A1**

29.

5(a)	charge / potential (difference)	**M1**
	charge is charge on one plate, <u>and</u> potential is p.d. across the plates	**A1**
5(b)	p.d. across both capacitors = E	**B1**
	$Q_T = Q_1 + Q_2$	**B1**
	$C_TE = C_1E + C_2E$ hence $C_T = C_1 + C_2$	**B1**
5(c)(i)	$[(1/22) + (1/47)]^{-1} = 15\ \mu F$	**A1**
5(c)(ii)	energy $= \frac{1}{2}CV^2$	**C1**
	$= \frac{1}{2} \times 15 \times 10^{-6} \times 12^2$ $= 1.1 \times 10^{-3}$ J	**A1**
5(c)(iii)	initial p.d. (across 22 μF) $= 12 \times (15 / 22)$ $= 8.2$ V **or** final p.d. across both capacitors $= 6.0 \times (22 / 15)$ $= 8.8$ V	**C1**

	$V = V_0 \exp [-t / (2.7 \times 10^6 \times 15 \times 10^{-6})]$	C1
	$6.0 = 8.2 \exp [-t / (2.7 \times 10^6 \times 15 \times 10^{-6})]$ **or** $8.8 = 12 \exp [-t / (2.7 \times 10^6 \times 15 \times 10^{-6})]$ $t = 13$ s	A1

30.

5(a)(i)	$Q = CV$	C1
	$Q_0 = 24 \times 470 \times 10^{-6}$ $= 0.011$ C	A1
5(a)(ii)	$I_0 = 24 / 5600$ $= 4.3 \times 10^{-3}$ A	A1
5(a)(iii)	$\tau = RC$	C1
	$= 5600 \times 470 \times 10^{-6}$ $= 2.6$ s	A1
5(a)(iv)	line with negative gradient throughout passing through $(0, I_0)$	B1
	exponential decay curve asymptotic to t-axis	B1
5(b)(i)	current in wire P gives rise to a magnetic field	B1
	as current (in P) changes, wire Q cuts (magnetic) flux (of wire P)	B1
	cutting magnetic flux causes induced e.m.f. (across Q)	B1
5(b)(ii)	sketch shows line with a negative gradient throughout	B1

31.

6(a)	• p.d. across resistor = p.d. across capacitor • current (in resistor) proportional to p.d. across it • current causes capacitor to lose charge • charge (on capacitor) proportional to p.d. so p.d. decreases *Any two points, 1 mark each*	B2
	rate of change of p.d. decreases as p.d. decreases	B1
6(b)	$Q_0 = 0.90$ mC **and** at t = one time constant, $Q = Q_0 \exp(-1)$	B1
	at t = one time constant, $Q = 0.90 \exp(-1) = 0.33$ mC	M1
	evidence of graph reading: when $Q = 0.33$ mC, $t = 5.5$ s	A1
	or	
	evidence of two correct sets of readings for Q and t from the graph	(B1)
	correct substitution of Q and t values into $Q_2 = Q_1 \exp [(t_1 - t_2) / \tau]$	(M1)
	calculation to give $\tau = 5.5$ s	(A1)
	or	
	read-off of half-life as 3.75 s	(B1)
	use of $Q = Q_0 \exp(-t / \tau)$ to show that τ = half-life / ln 2	(M1)
	$\tau = 3.75 / \ln 2 = 5.4$ s	(A1)

6(b)	**or**	
	tangent drawn on Q–t graph and value of Q at exact same time as tangent read from graph	**(M1)**
	gradient of tangent correctly calculated	**(A1)**
	$\tau = Q$ / gradient used to correctly calculate a value for τ as 5.5 s	**(A1)**
6(c)(i)	$C = Q / V$	**C1**
	$= [(0.90 \times 10^{-3}) / 7.5] = 1.2 \times 10^{-4}$ C $= 120\ \mu$F	**A1**
6(c)(ii)	$R = \tau / C$	**C1**
	$= 5.5 / (1.2 \times 10^{-4})$ $(= 45\,800\ \Omega)$ $= 46$ kΩ	**A1**

32.

5(a)	from graph ln $Q = 2.9$ (so $Q = 18.2\ \mu$C)	**B1**
	$C = Q / V$	**C1**
	$= 18.2 / 12 = 1.5\ \mu$F	**A1**
5(b)	gradient $= -0.25$	**C1**
	gradient $= -1 / RC$	**C1**
	$R = 1 / (0.25 \times 1.5 \times 10^{-6})$ $= 2.7 \times 10^{6}\ \Omega$	**A1**
	or $\frac{Q}{Q_0} = e^{-t/CR}$ **or** $\ln Q - \ln Q_0 = \frac{-t}{CR}$	**(C1)**
	e.g. $\frac{4.95}{18.2} = e^{-5.2/(1.5 \times 10^{-6} R)}$ **or** $1.6 - 2.9 = 5.2 / (1.5 \times 10^{-6} R)$	**(C1)**
	$R = 2.7 \times 10^{6}\ \Omega$	**(A1)**
5(c)	$W = ½\ QV$	**C1**
	$= ½ \times 18.2 \times 10^{-6} \times 12$ $= 1.1 \times 10^{-4}$ J	**A1**
	or $W = ½\ CV^2$	**(C1)**
	$= ½ \times 1.5 \times 10^{-6} \times 12^2$ $= 1.1 \times 10^{-4}$ J	**(A1)**
	or $W = ½\ Q^2 / C$	**(C1)**
	$= ½ \times (18.2 \times 10^{-6})^2 / 1.5 \times 10^{-6}$ $= 1.1 \times 10^{-4}$ J	**(A1)**
5(d)	straight line with different negative gradient starting from (0, 2.9)	**M1**
	straight line between $t = 0$ and at least $t = 5.0$ s with twice the gradient of the original line	**A1**

33.

5(a)(i)	$Q_A = CV$	**A1**
5(a)(ii)	$E_A = ½CV^2$	**A1**
5(b)(i)	some of the charge transfers to (the plates of) capacitor B	**B1**
	transfer is because the p.d.s across the capacitors are not equal **or** transfer stops when the p.d.s across the capacitors become equal	**B1**
5(b)(ii)	$V_A = V_B$	**M1**
	charge on A + charge on B = CV	**M1**
	$CV_B + 3CV_B = CV$ leading to $V_B = V/4$	**A1**
	or	
	$C_T = 4C$	**(M1)**
	$Q_T = CV$	**(M1)**
	$V_B = CV/4C = V/4$	**(A1)**
5(b)(iii)	$\Delta E = ½CV^2 - nCV^2$, where n is a multiple that is less than ½ **or** total final energy $= ½ \times 4C \times (V/4)^2$ $= ⅛CV^2$	**C1**
	$\Delta E = ½CV^2 - ⅛CV^2$ $= ⅜CV^2$	**A1**

34.

6(a)(i)	energy stored = area under graph	**C1**
	$= ½ \times 450 \times 10^{-6} \times 8.0 = 1.8 \times 10^{-3}$ J **or** 1.8 mJ	**A1**
6(a)(ii)	$C = Q/V$ **or** $E = ½CV^2$	**C1**
	$C = (450 \times 10^{-6}) / 8.0$ **or** $(2 \times 1.8 \times 10^{-3}) / 8.0^2$ $= 5.6 \times 10^{-5}$ F	**A1**
6(b)(i)	$V = V_0 \exp(-t/RC)$ **and** $\tau = RC$	**C1**
	$V = V_0 \exp(-t/\tau)$ $V_0 = 8.0$ V, and at one time constant, $t = \tau$ $V/8.0 = \exp(-\tau/\tau)$, so $\ln(V/8.0) = -1.0$ **or** $-\ln(V/8.0) = 1.0$	**A1**
6(b)(ii)	[t read from graph at $-\ln(V/8.0) = 1.0$]: $\tau = 3.2$ s	**A1**
6(b)(iii)	$\tau = RC$	**C1**
	$R = 3.2 / (5.6 \times 10^{-5})$ $= 5.7 \times 10^4\ \Omega$	**A1**